



DEEP DIVE WORKSHOP

INTRODUCTION TO **Vector** Embeddings



ZION PIBOWEI
AI Dev Experience Manager

Agenda

- 1 Introduction
- 2 Motivation
- 3 What really are embeddings?
- 4 How are embeddings learned?

Introduction



It's scary how well @Spotify Discover Weekly playlists know me. Like former-lover-who-lived-through-a-near-death experience-with-me well.

8:09 PM · Oct 27, 2015



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Amanda Whitbred
@amandawhitbred · Follow



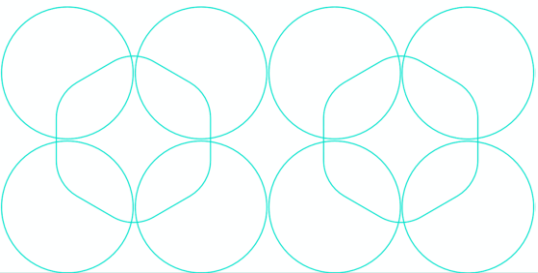
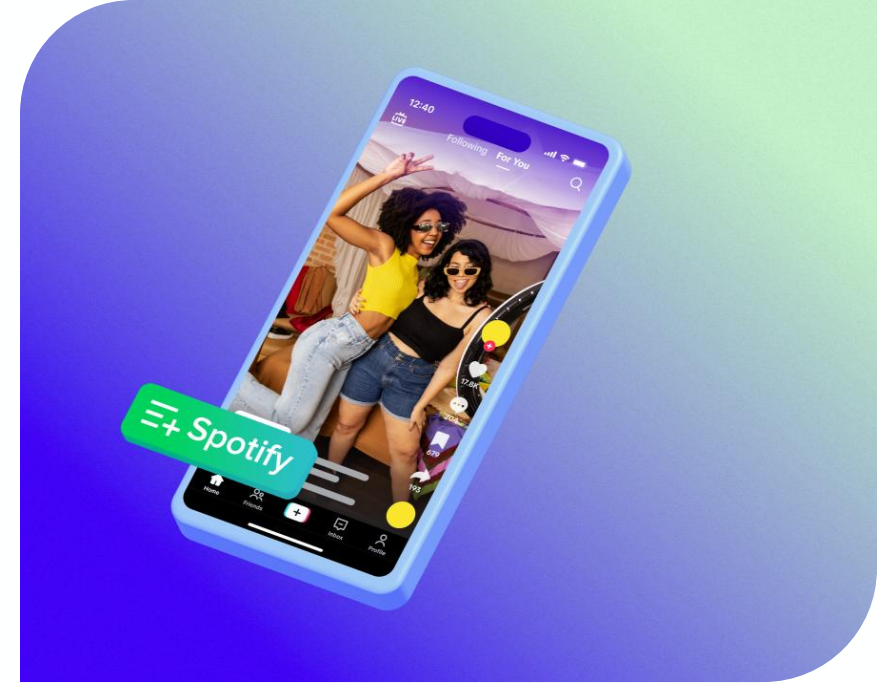
At this point @Spotify's discover weekly knows me so well that if it proposed I'd say yes

10:36 PM · Aug 18, 2016



❤️ 76 💬 Reply 🔗 Copy link

Read 1 reply



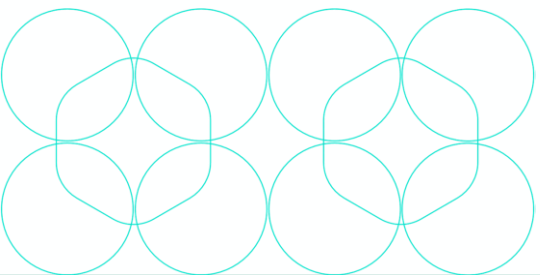
Introduction



How exactly does
Spotify know you will
like a song you've never
heard?

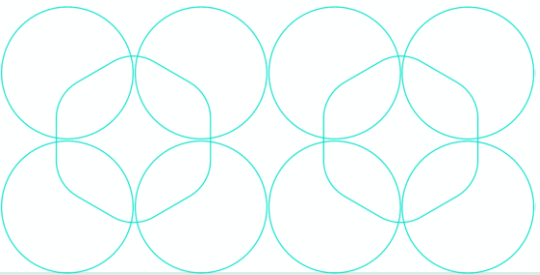
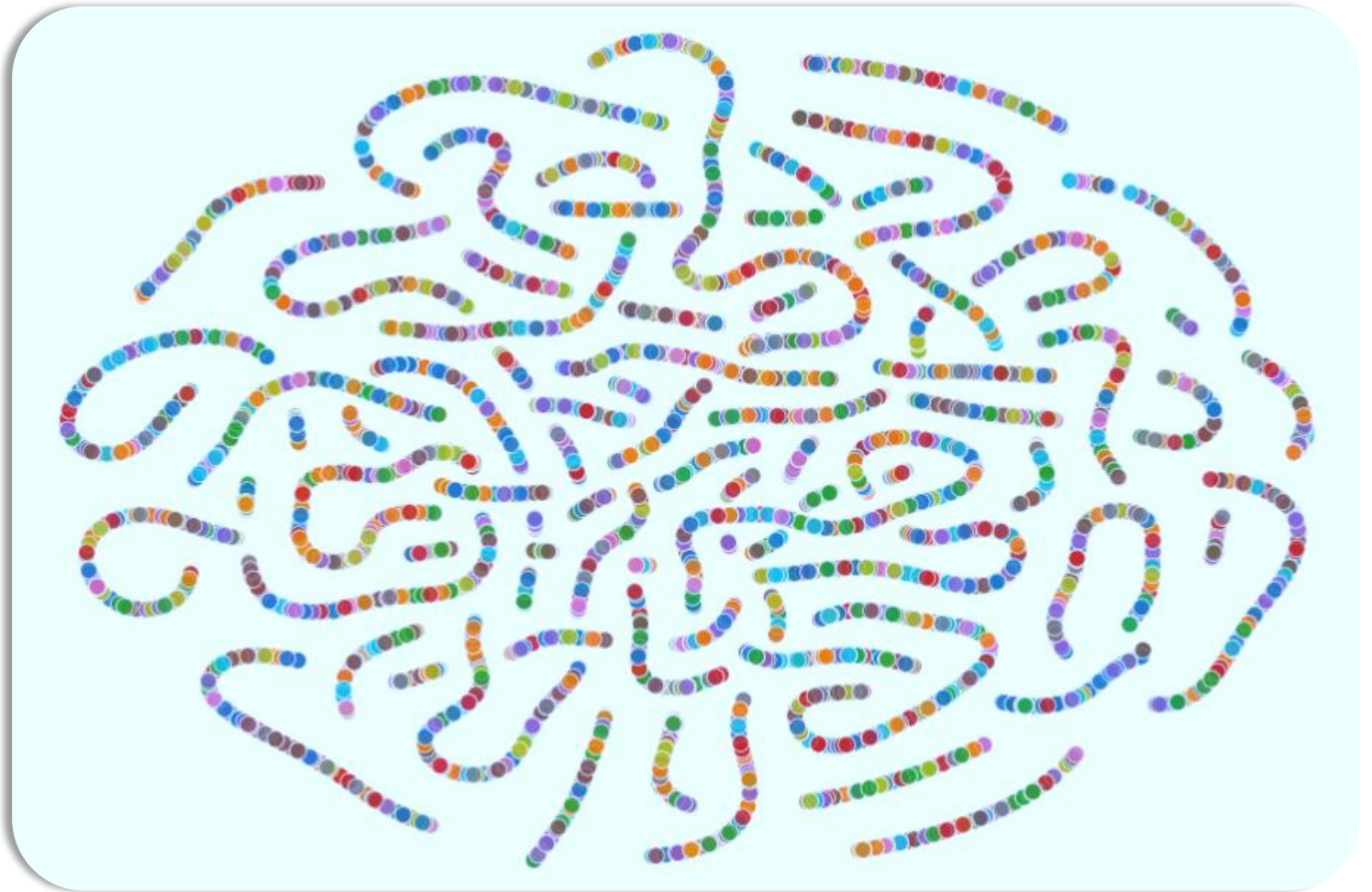
Spotify solves the problem of
choice overload by using powerful
recommendation algorithms that
act as your musical compass.

The shuffle feature seems to magically
play just the right song at the right
moment. What sorcery is that? 🤖



Motivation

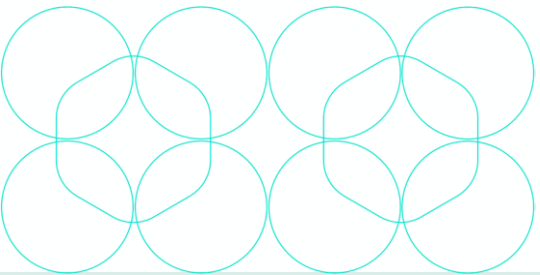
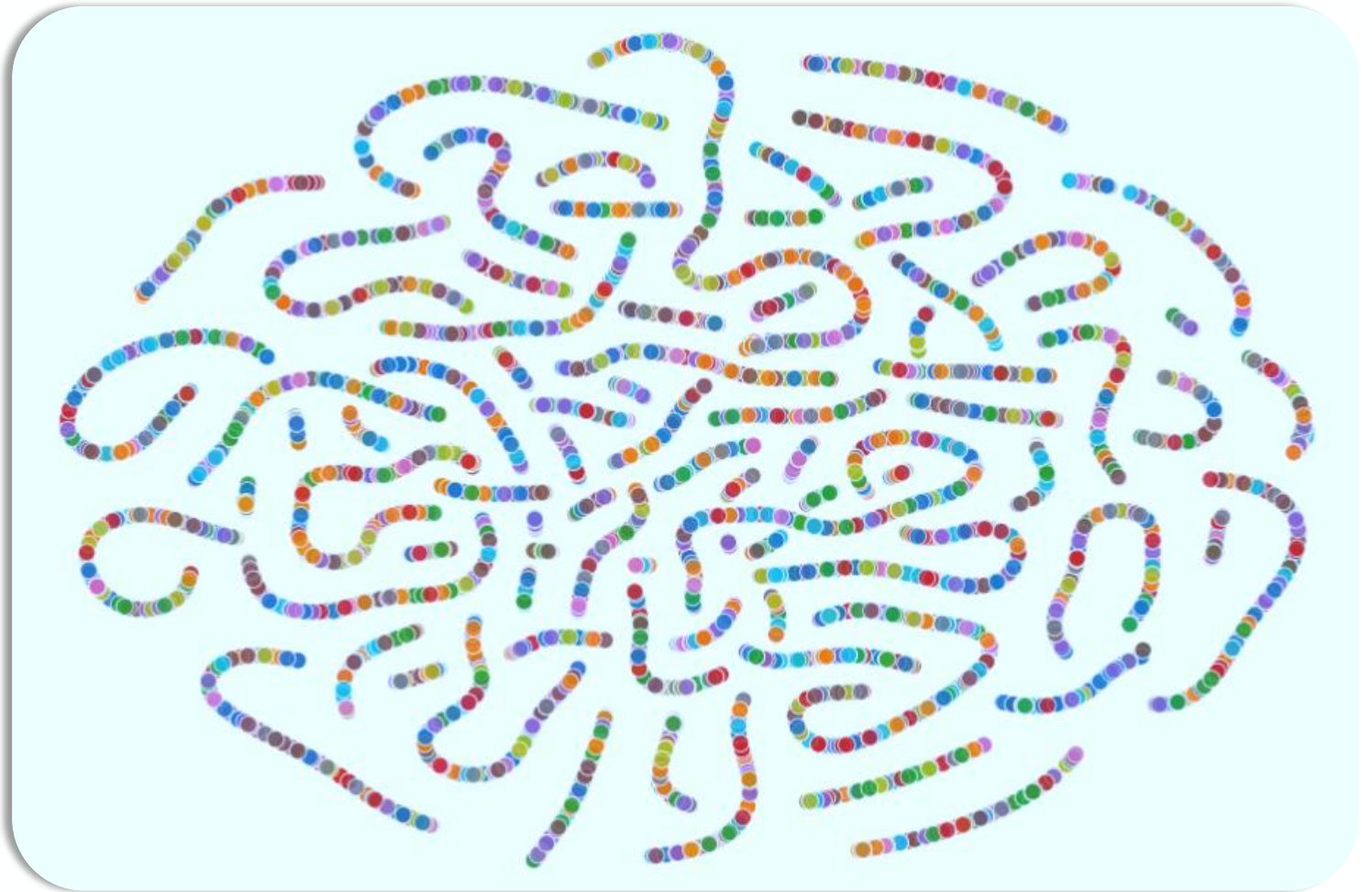
Let's start with a simple
music **cluster**



Motivation

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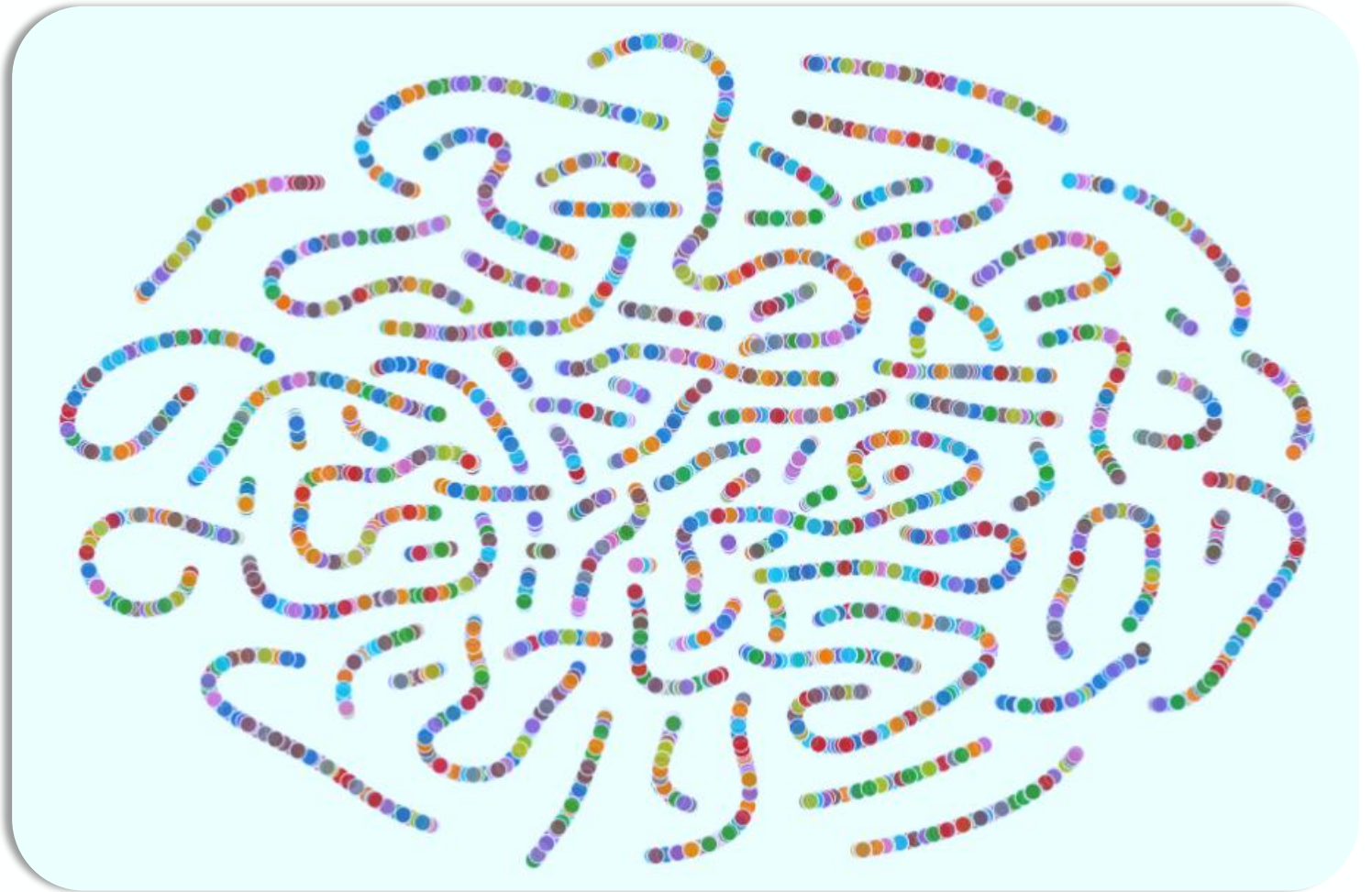
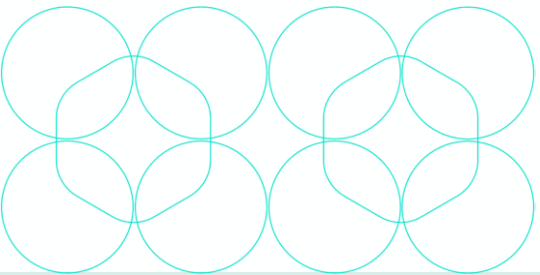
This is generated using the **t-distributed Stochastic Neighbor Embedding (t-SNE)** algorithm



Motivation

Steps for generating the cluster

- Download the Spotify dataset from [Kaggle](#)
- Load the data into a Pandas dataframe
- Select specific numerical features that quantify different audio attributes
- Apply t-SNE algorithm to learn embeddings from these audio features
- Plot clusters of similar music.
- Voila!



Motivation

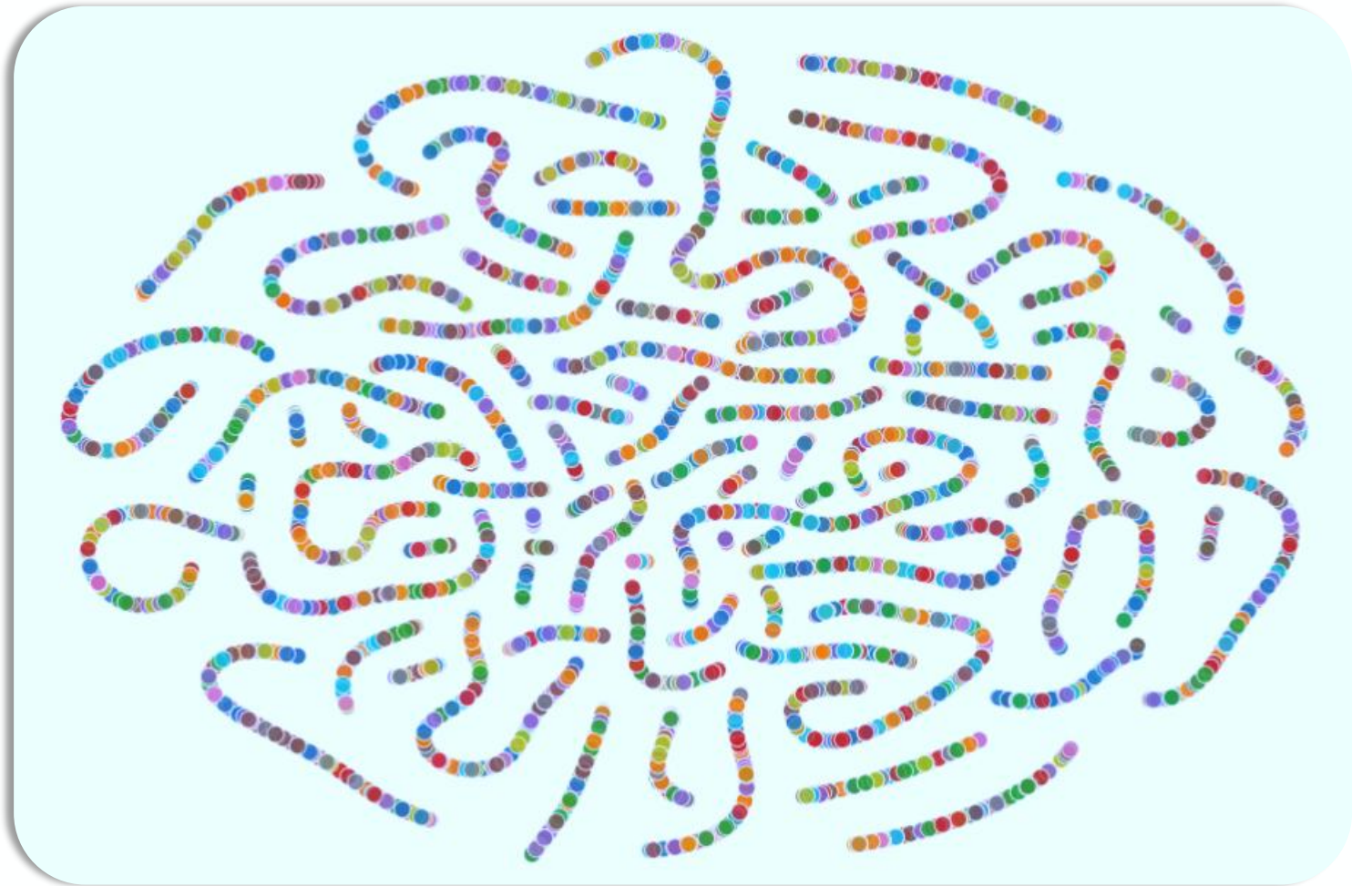
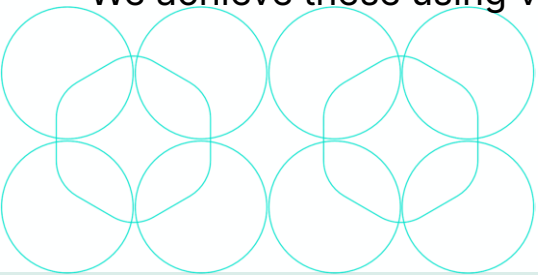
What is the essential concept here?

One word: Representation

Representation of complex things (music, images, words) as points in a high-dimensional space.

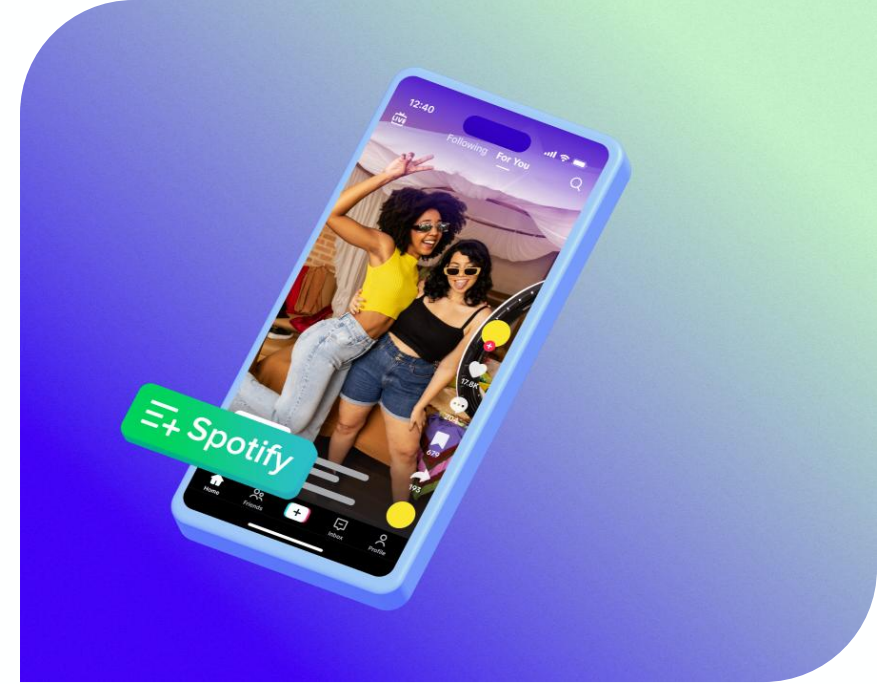
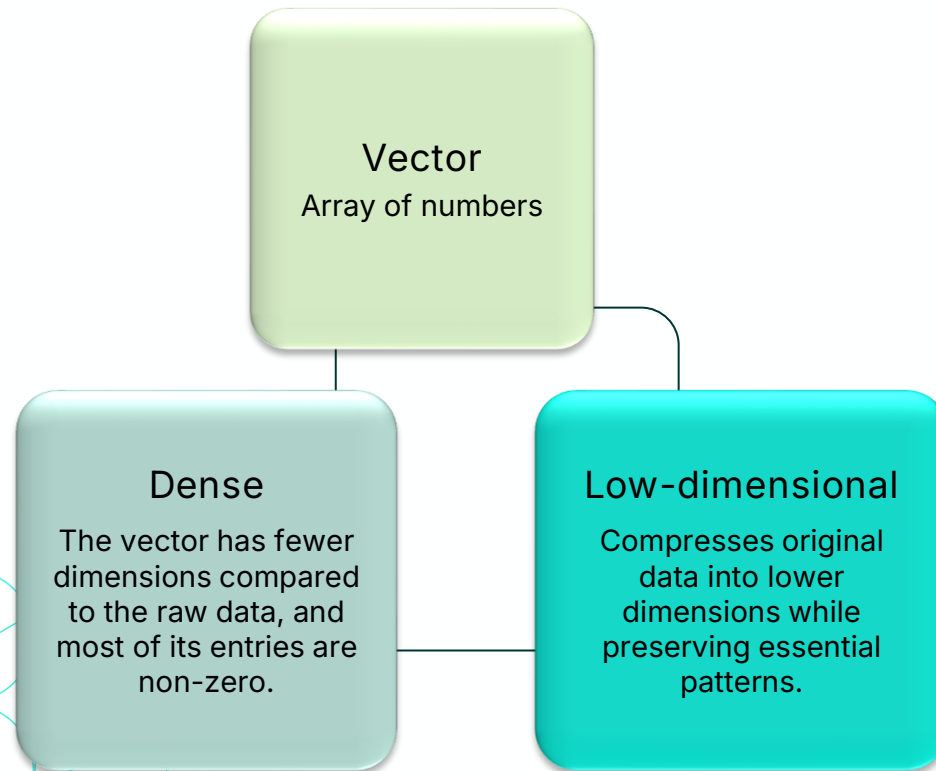
This enables us to quantify similarities, uncover semantic relationships, and create magical recommendations

We achieve these using vector embeddings.



So, what really are embeddings?

Embeddings are dense, low-dimensional vector representations of data where similar things are close together.

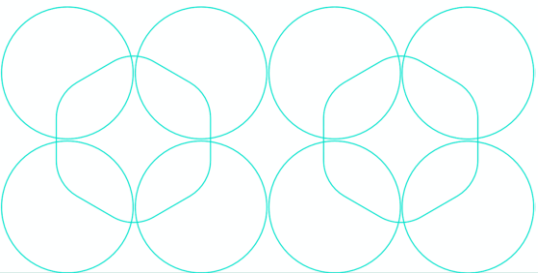
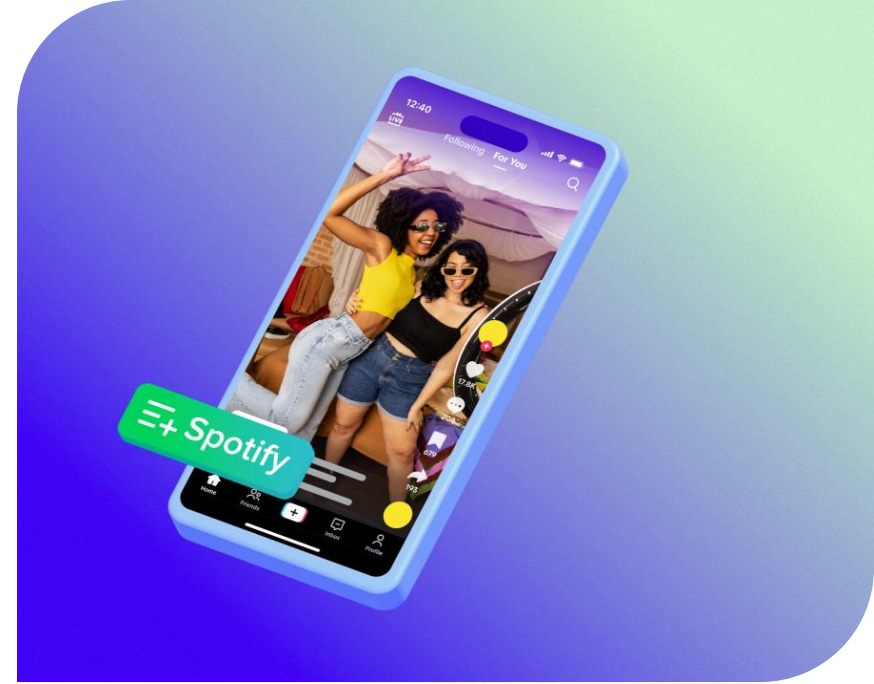


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Key Properties

- Captures similarity: Items with similar meanings or properties have vectors that are close to each other
- Encodes semantic relationships: "King" and "Queen" would have vectors close to each other.
- Makes data usable for algorithms: Computations (like similarity searches) become faster and more meaningful.

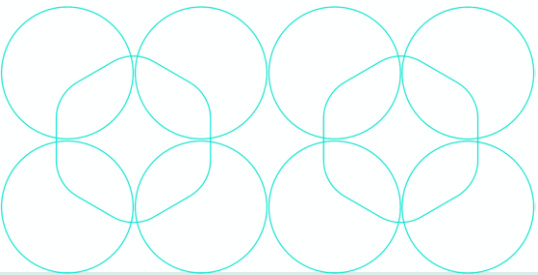
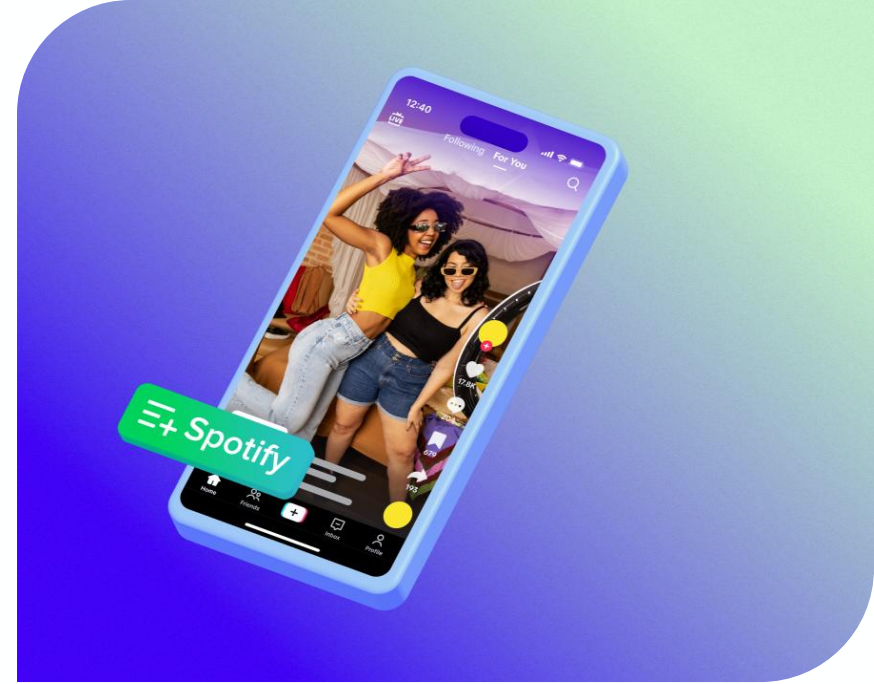


How are embeddings learned?

Models learn embeddings by optimizing a task, such as predicting context words in Word2Vec, reconstructing input in autoencoders, etc.

Optimization Process:

- Models (like neural networks) start with random embeddings for items (e.g., words, songs, images).
- During training, the model adjusts these vectors to minimize a loss function (a measure of error).
- Over many iterations, embeddings evolve so that items with similar roles or meanings end up close together in the embedding space.

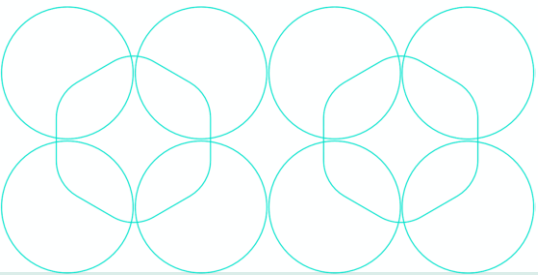
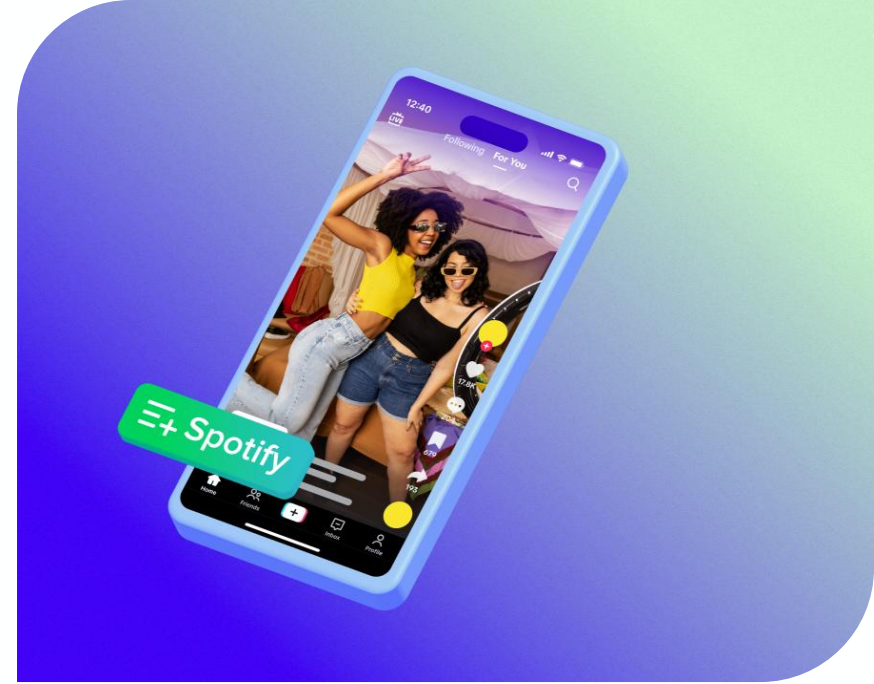


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Example: Word embeddings

- Examples: Word2Vec, GloVe, etc
- Task: Predict missing words in sentences (e.g., "The cat sat on the ____").
- Process: The neural network learns word vectors that make it easier to predict correct words.
- Result: Words that appear in similar contexts (like "cat" and "dog") end up with similar vectors.

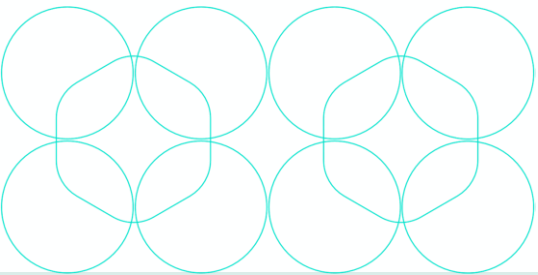
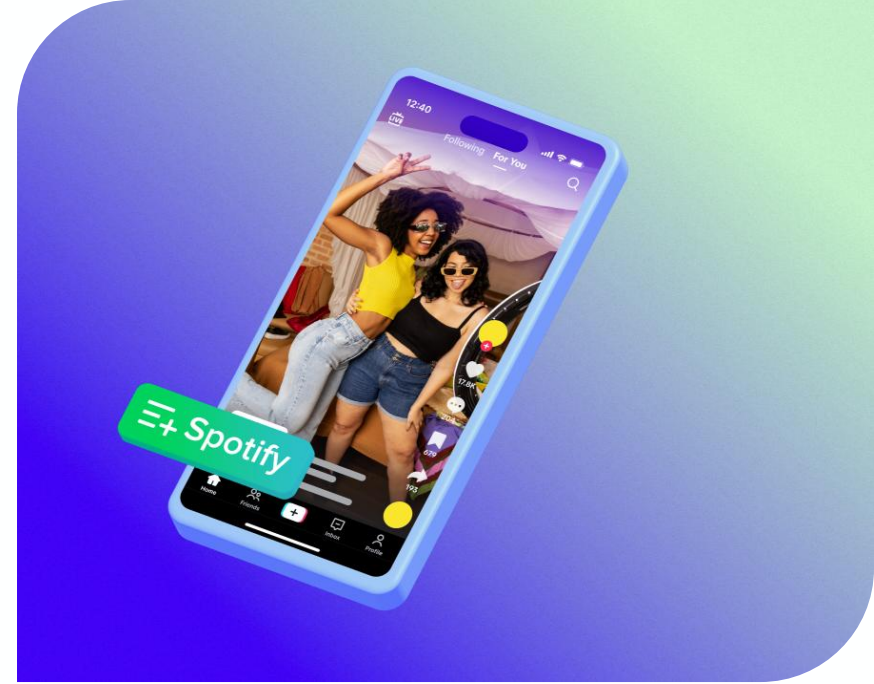


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Example: Item embeddings

- Examples: Songs in Spotify, Products in Amazon
- Task: Predict user preferences (e.g., recommend songs/products).
- Process: The model learns embeddings that capture patterns of co-listening or co-purchasing.
- Result: Songs often played together end up close in embedding space.

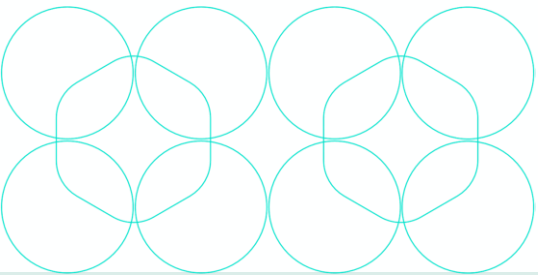
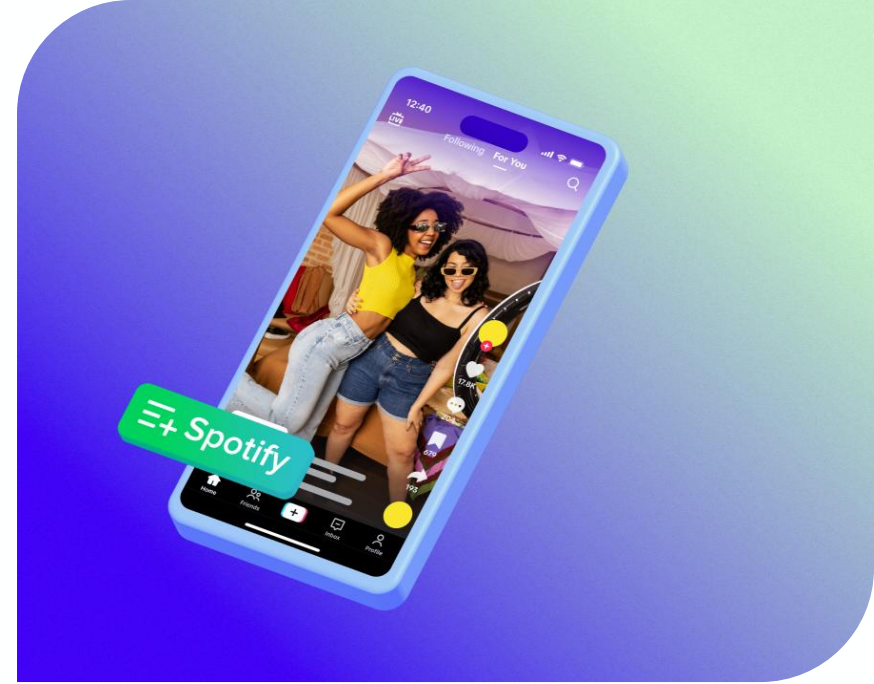


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Example: Image embeddings

- Task: Classify images into categories such as cats, dogs, cars, etc.
- Process: The neural network learns to transform raw pixels into embeddings where images of cats cluster together.
- Result: Even without explicit labels, embeddings capture visual similarity.



Let's Code!

 Open in Colab

